



JURONG JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2018

CANDIDATE
NAME

Mark Scheme

CLASS

18S

EXAM INDEX

CHEMISTRY

9729/04

Higher 2

Paper 4 Practical

16 August 2018
2 hours 30 minutes

Candidates answer on the Question paper.

Additional Materials: As listed in the Confidential Instructions

READ THESE INSTRUCTIONS FIRST

Write your name, class and exam index number on all the work you hand in.
Give details of the practical shift and laboratory where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use a HB pencil for any diagrams, graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.
Qualitative Analysis Notes are printed on pages 16 and 17.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Shift	
Laboratory	

Mark Scheme:

For calculations, the principle of no double penalty (error carried forward) applies.

For connecting parts, marking from point of first penalty onwards will be based on correct method only.

For Examiner's Use	
1	13
2	16
3	12
4	14
Total	55

This document consists of **17** printed pages and **0** blank page.

Answer **all** the questions in the spaces provided.

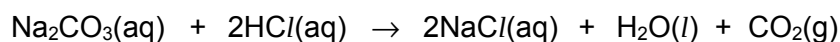
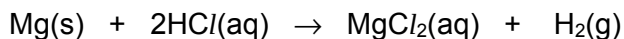
For
Examiner's
Use

1 Determination of the A_r of magnesium by a back titration method

FA 1 is an aqueous solution prepared by dissolving 0.215 g of magnesium ribbon in 30.0 cm³ of 1.00 mol dm⁻³ hydrochloric and made up to 250 cm³ with deionised water.

FA 2 is aqueous sodium carbonate containing 2.65 g dm⁻³ Na₂CO₃.

In this question, you will titrate **FA 2** with **FA 1** to determine how much hydrochloric acid was left over after the reaction with magnesium.



Using the experimental results, you will determine the A_r of magnesium.

(a) Method

1. Fill the burette with **FA 1**.
2. Using a pipette, transfer 25.0 cm³ of **FA 2** into the conical flask.
3. Add a few drops of methyl orange indicator.
4. Run **FA 1** from the burette into the conical flask until the yellow colour of the solution changes to orange.
5. Record your titration results in the space provided below. Make certain that your recorded results show the precision of your working.
6. Repeat points 1 to 5 as necessary until consistent results are obtained.

Results

Titration number	1	2
Final burette reading / cm ³	23.80	47.80
Initial burette reading / cm ³	0.00	24.00
Volume of FA 1 used / cm ³	23.80	23.80

✓

✓

[1] A proper table with correct headers (must state '*burette*') and units. (Do not award if any final and initial burette readings are inverted)

[1] All burette readings are recorded to the nearest 0.05 cm³ + correct computation of titres.

(Do not award if 50 is used as initial burette reading or burette reading is > 50)

[1] Has two uncorrected titres within 0.10 cm³ + place ✓ under selected titres within 0.10 cm³. (Do not award if additional titration is performed after consistent titres are obtained.)

Accuracy: Supervisor's titre, $V = 23.80 \text{ cm}^3$

$(V \pm 0.20) \text{ cm}^3$, 23.60 – 24.00 cm³ [2]

$(V \pm 0.40) \text{ cm}^3$, 23.40 – 24.20 cm³ [1]

[5]

- (b) From your titrations, obtain a suitable volume of **FA 1** to be used in your calculations. Show clearly how you obtained this volume.

$$\text{average volume of FA 1 used} = \frac{23.80 + 23.80}{2} = 23.80 \text{ cm}^3$$

- quotes the selected titres within 0.10 cm³ in the working
- calculates average titre correctly to 2 d.p. [1]

25.0 cm³ of **FA 2** requires 23.80 cm³ of **FA 1** for complete reaction. [1]

(c) Calculations

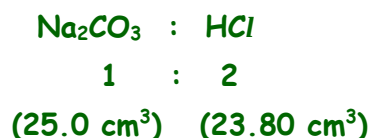
- (i) Calculate the amount of sodium carbonate in the 25.0 cm³ of **FA 2** used in each titration. [*A_r*: Na, 23.0; C, 12.0; O, 16.0]

$$\text{Amount of Na}_2\text{CO}_3 = \frac{2.65}{106.0} \times \frac{25.0}{1000} = 6.25 \times 10^{-4} \text{ mol} \quad [1]$$

Amount of Na₂CO₃ in 25.0 cm³ of **FA 2** = 6.25 × 10⁻⁴ mol [1]

- (ii) Use your answer in (c)(i) to calculate the amount of hydrochloric acid present in 250 cm³ of **FA 1**.

$$\begin{aligned} \text{Amount of HCl in 250 cm}^3 \\ = \frac{250}{\text{ans. (b)}} \times (2 \times \text{ans. (c)(i)}) \end{aligned}$$



$$= \frac{250}{23.80} \times (2 \times 6.25 \times 10^{-4}) = 0.01313 \text{ mol} \quad [1]$$

[1]

Amount of HCl present in 250 cm³ = 0.0131 mol

- (iii) Calculate the amount of hydrochloric acid that reacted with the magnesium.

Amount of HCl reacted with Mg

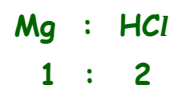
$$\begin{aligned} &= \left(\frac{30}{1000} \times 1.00 \right) - \text{ans. (c)(ii)} \\ &= \left(\frac{30}{1000} \times 1.00 \right) - 0.01313 = 0.01687 \text{ mol} \quad [1] \end{aligned}$$

Amount of HCl that reacted with the magnesium = 0.0169 mol [1]

- (iv) Hence calculate the relative atomic mass, *A_r*, of magnesium.

A_r of Mg

$$= \frac{0.215}{\left(\frac{1}{2} \times \text{ans. (c)(iii)} \right)} = \frac{0.215}{\left(\frac{1}{2} \times 0.01687 \right)} = 25.49 \quad [1]$$



A_r of magnesium = 25.5 (no units) [1]

- (d) The A_r of magnesium given in the Periodic Table is 24.3. Use this value given in the Periodic Table to calculate the percentage error of your A_r value of Mg obtained in (c)(iv).

$$\begin{aligned}\% \text{ error} &= \frac{|\text{ans(c)(iv)} - 24.3|}{24.3} \times 100\% \\ &= \frac{|25.49 - 24.3|}{24.3} \times 100\% = 4.90\% \quad [1]\end{aligned}$$

$$\% \text{ error} = \underline{4.90\%} \quad [1]$$

- (e) A solution of sodium hydroxide was prepared at the same concentration, in mol dm^{-3} , as FA 2. A student repeated the titration but replaced FA 2 with this sodium hydroxide.

Explain the effect that replacing FA 2 with this solution of sodium hydroxide would have on the volume of FA 1 needed for the titration.

Half the volume of FA 1 is required since NaOH and HCl react in a mole ratio of 1:1 while Na_2CO_3 and HCl react in a mole ratio of 1:2.

[1]

[2]

[1] State correct units in 1(b)[cm^3], (c)(i)(ii)(iii)[mol], (c)(iv)[no units] and 1(d)[%]

Any calculation not attempted loses this mark.

[Total: 13]

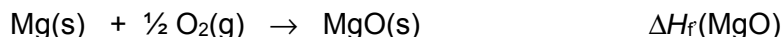
2 Determination of enthalpy change of formation of magnesium oxide.

FA 3 is magnesium ribbon

FA 4 is magnesium oxide

FA 5 is 2.0 mol dm⁻³ hydrochloric acid.

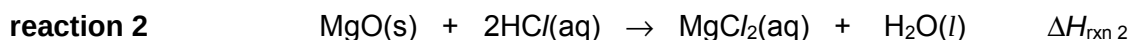
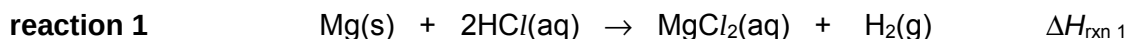
The enthalpy change of formation of magnesium oxide, $\Delta H_f(\text{MgO})$, is the enthalpy change when one mole of magnesium oxide is formed from its elements in their standard states under standard conditions.



To cause magnesium and oxygen to react together, they must be heated. The reaction then proceeds in an uncontrolled and highly exothermic fashion. It is not possible to measure this enthalpy change without the use of equipment such as a bomb calorimeter.

You are to perform an experiment by which you will determine this enthalpy change by an indirect route.

Both magnesium and magnesium oxide react exothermically with hydrochloric acid to form magnesium chloride. You will determine the molar enthalpy changes for **reaction 1** and **reaction 2**.



You will then use Hess's Law to determine a value for the enthalpy change of formation of magnesium oxide, $\Delta H_f(\text{MgO})$.

(a) Determining the enthalpy change of reaction 1 between FA 3 and FA 5.

Follow the instructions below to determine the maximum temperature change when a known mass of magnesium, **FA 3**, reacts completely with excess **FA 5**. Record all the mass readings in Table 2.1 and all measurements of time and temperature in the space provided below the table.

Method 1

1. Weigh the stoppered weighing bottle labelled **FA 3** and its contents. Record the total mass of weighing bottle and **FA 3**.
2. Use a 50 cm³ measuring cylinder to transfer 50.0 cm³ of **FA 5** into a Styrofoam cup. Place this cup inside a second Styrofoam cup which is held in a 250 cm³ glass beaker to prevent it tipping over.
3. Stir the solution in the cup with the thermometer and measure its temperature. Record this temperature in your table in **(a)(i)** (this is the temperature at $t = 0$ min) and start the stop-watch. Repeat this measurement each minute for 3 minutes and record each temperature.
4. At the fourth minute, carefully add **FA 3** from the weighing bottle to **FA 5** in the cup. Quickly stir the mixture with the thermometer but **do not measure the temperature**.
5. Continue to stir the mixture. Measure and record the temperature at 4½ minutes, then **every half minute** until the temperature reaches a maximum, and then **every minute** until the tenth minute.
6. Empty and wash the Styrofoam cup. Rinse the cup with deionised water and dry it using a paper towel.
7. Reweigh the emptied weighing bottle and record its mass. Calculate and record the mass of **FA 3** used.

(a) (i) Results

Record the required mass readings in Table 2.1 and in an appropriate format in the space below, record all measurements of time and temperature.

Table 2.1

Mass of weighing bottle with FA 3 /g	5.382
Mass of empty weighing bottle / g	5.248
Mass of FA 3 used / g	0.134

Measurements of time and temperature:

Time / min	Temperature / °C
0	29.6
1	29.6
2	29.6
3	29.6
4	-
4½	41.6
5	42.0
5½	41.6
6	41.3
7	40.6
8	40.0
9	39.4
10	38.9

[1] Tabulates all time-temp data in a single table with correct headers and units. If headers have no units, then each reading must have units stated + records all masses consistently to 2/3 d.p. and correctly calculates mass of FA3.

[1] Records all time readings consistently in the same format e.g. "4½" "5" "5½" / "4.5" "5.0" "5.5" / "4 min 30 s" "5 min 0 s" + changes from ½ min to 1 min intervals AFTER maximum temp reached + need to have at least one ½ minute reading after the maximum temp.

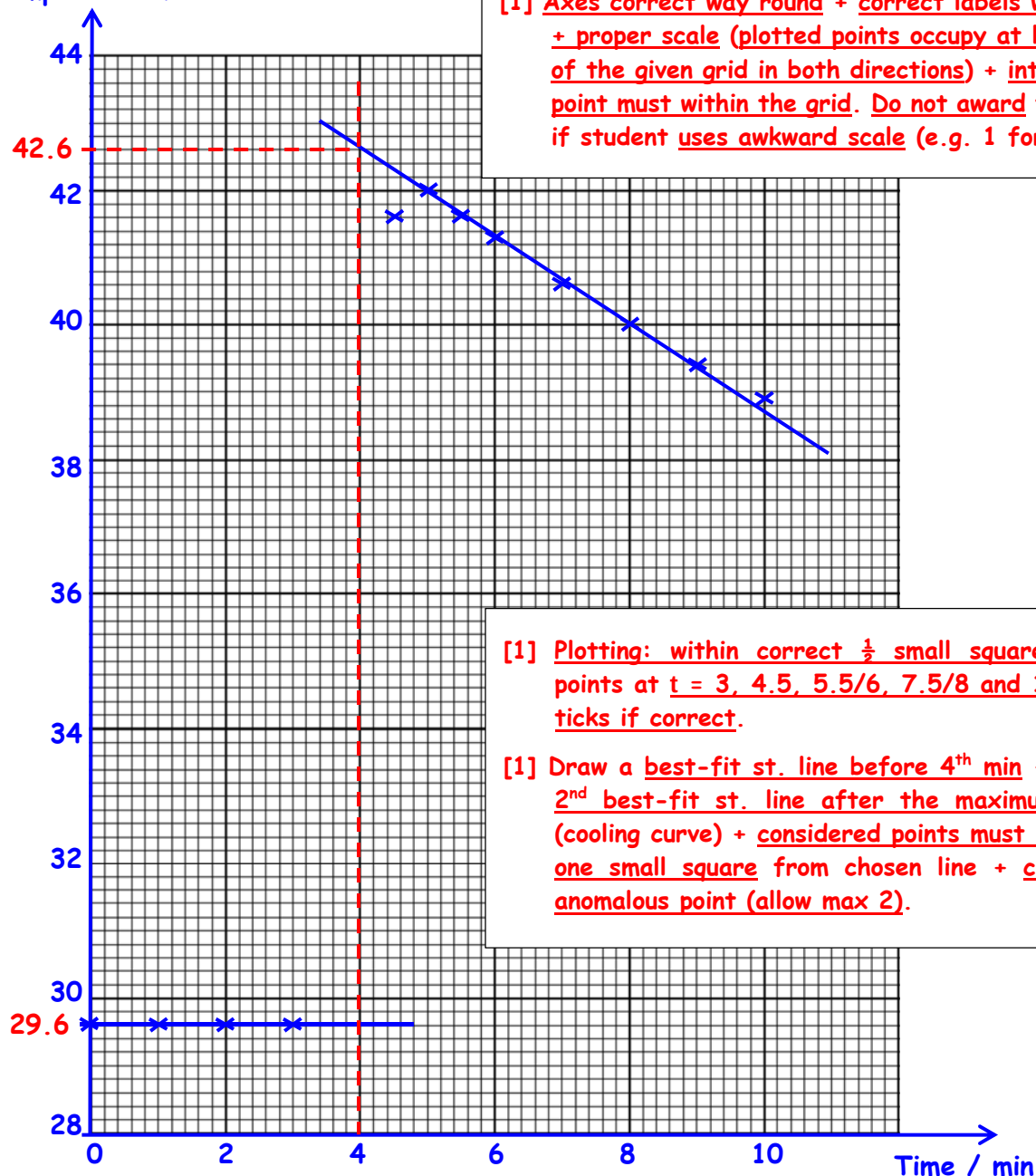
[1] Records ALL temperature readings to the nearest 0.1 °C (1 d.p.) (using 0.2 °C interval thermometer)

[3]

- (a) (ii) Plot a graph of temperature (y-axis) against time (x-axis) on the grid below. You will use the graph to determine the theoretical temperature change at 4 minutes. The scale for temperature should extend at least 1°C above your highest recorded temperature.

1. Draw a straight line of best-fit for the points before the fourth minute. Draw a second straight line of best-fit for the points after the fourth minute. Extrapolate both lines to the fourth minute.
2. From your graph, read the minimum and maximum temperature at the fourth minute. Record these values in the spaces below and determine a value for the maximum temperature rise, ΔT , at the fourth minute.

Temperature / $^{\circ}\text{C}$



[1] Axes correct way round + correct labels with units + proper scale (plotted points occupy at least half of the given grid in both directions) + intersection point must within the grid. Do not award this mark if student uses awkward scale (e.g. 1 for 3/7)

[1] Plotting: within correct $\frac{1}{2}$ small square. Check points at $t = 3, 4.5, 5.5/6, 7.5/8$ and 10 ; place ticks if correct.

[1] Draw a best-fit st. line before 4th min + draw a 2nd best-fit st. line after the maximum temp. (cooling curve) + considered points must be within one small square from chosen line + circle any anomalous point (allow max 2).

[1] Extrapolates the both straight lines to intersect the vertical line (line may not be drawn) at 4 min + states min temp, max temp and ΔT correctly to the nearest half square or in 1 d.p. (following the precision of thermometer)

Temperature at 4th minute
 minimum = 29.6°C
 maximum = 42.6°C
 temperature rise, $\Delta T = 13.0^{\circ}\text{C}$ [4]

- (iii) Use your ΔT from (a)(ii) to calculate a value for the molar enthalpy change of **reaction 1** between **FA 3** and **FA 5**, $\Delta H_{\text{rxn 1}}$, in kJ mol^{-1} . Assume that the specific heat capacity of the final solution is $3.75 \text{ J g}^{-1} \text{ K}^{-1}$ and that its density is 1.05 g cm^{-3} . [A_r: Mg, 24.3]

$$\begin{aligned}\text{Heat evolved} &= q = mc\Delta T \\ &= (50.0 \times 1.05) \times 3.75 \times 13.0 \text{ J} \\ &= 2559 \text{ J} = 2.559 \text{ kJ} \text{ [1] (in J or kJ) ignore sign}\end{aligned}$$



Mg is limiting, HCl is in excess.

$$\Delta H_{\text{rxn 1}} = -\frac{2.559}{\left(\frac{0.134}{24.3}\right)} = -464 \text{ kJ mol}^{-1} \text{ [1] with -ve sign}$$

$$\Delta H_{\text{rxn 1}} = \underline{-464} \text{ kJ mol}^{-1} \text{ [2]}$$

(b) Determining the enthalpy change of reaction 2 between FA 4 and FA 5.

A student carried out an experiment to determine the enthalpy change of **reaction 2**, $\Delta H_{\text{rxn } 2}$, using a known mass of **FA 4** that reacts with excess **FA 5**.

The student was provided with five pre-weighed samples of **FA 4**, labelled **v**, **w**, **x**, **y** and **z**, and carried out the experiment according to the following instructions.

Method 2

1. Use a 50 cm³ measuring cylinder to transfer 50.0 cm³ of **FA 5** into a Styrofoam cup. Place this cup inside a second Styrofoam cup which is held in a 250 cm³ glass beaker to prevent it tipping over.
2. Stir the solution in the cup with the thermometer. Read and record its temperature. This is the initial temperature of **FA 4**, T_{initial} .
3. Carefully add sample **v** to the **FA 5** in the cup.
4. Using the thermometer, stir the mixture continuously until it reaches its maximum temperature. Record the maximum temperature reached, T_{max} .
5. Wash and carefully dry the Styrofoam cup.
6. Repeat steps 1 to 5 with the remaining samples **w**, **x**, **y** and **z**.

The student recorded all masses and temperatures in Table 2.2 below.

Table 2.2

	expt v	expt w	expt x	expt y	expt z
Mass of FA 4 added to the cup, m / g	0.507	0.503	0.487	0.506	0.476
T_{initial} / °C	31.2	31.2	31.0	31.1	31.3
T_{max} / °C	38.7	38.6	38.2	38.2	38.3
Temperature change, ΔT / °C	7.5	7.4	7.2	7.1	7.0
$\frac{\Delta T}{m}$ / °C g ⁻¹	14.8	14.7	14.8	14.0	14.7

- (i) Complete Table 2.2 by calculating values for ΔT and $\Delta T/m$. Leave values for $\Delta T/m$ to 3 significant figures.

[1] correctly computes ΔT to 1 d.p. and ' $\Delta T/m$ ' to 3 s.f.

- (ii) Based on the calculated $\Delta T/m$ values in Table 2.2, determine an appropriate mean maximum temperature change, ΔT_{ave} , and mean mass of **FA 4**, m_{ave} .

$$\Delta T_{\text{ave}} = \frac{7.5 + 7.4 + 7.2 + 7.0}{4} = 7.3 \text{ (1 d.p.)}$$

$$m_{\text{ave}} = \frac{0.507 + 0.503 + 0.487 + 0.476}{4} = 0.493 \text{ (3 d.p.)}$$

[1] correctly calculates both ΔT_{ave} (in 1 d.p.) and m_{ave} (in 3 d.p.) based on 4 consistent values (excludes one anomalous result)

$$\Delta T_{\text{ave}} = 7.3 \text{ }^{\circ}\text{C} \quad m_{\text{ave}} = 0.493 \text{ g} \quad [2]$$

- (iii) Use your results in (b)(ii) to calculate a value for the molar enthalpy change of **reaction 2** between **FA 4** and **FA 5**, $\Delta H_{\text{rxn } 2}$, in kJ mol^{-1} . Assume that the specific heat capacity of the final solution is $3.75 \text{ J g}^{-1} \text{ K}^{-1}$ and that its density is 1.05 g cm^{-3} . [A_r: Mg, 24.3; O, 16.0]

$$\begin{aligned}\text{Heat evolved} &= q = mc\Delta T_{\text{ave}} \\ &= (50.0 \times 1.05) \times 3.75 \times 7.3 \text{ J} \\ &= 1437 \text{ J} = 1.437 \text{ kJ} \text{ [1] (in J or kJ) ignore sign!}\end{aligned}$$

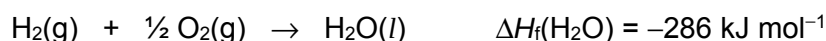


MgO is limiting, HCl is in excess.

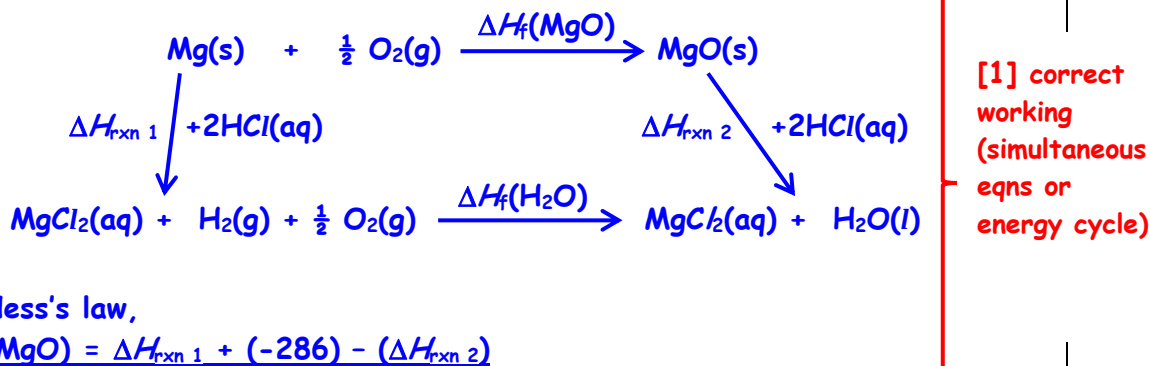
$$\Delta H_{\text{rxn } 2} = -\frac{1.437}{\left(\frac{0.493}{40.3}\right)} = -117 \text{ kJ mol}^{-1} \text{ [1] with -ve sign}$$

$$\Delta H_{\text{rxn } 2} = -117 \text{ kJ mol}^{-1} \text{ [2]}$$

- (c) Use your values for $\Delta H_{\text{rxn } 1}$ and $\Delta H_{\text{rxn } 2}$, together with the enthalpy change of formation of water, $\Delta H_f(\text{H}_2\text{O})$, to calculate a value for the enthalpy change of formation of magnesium oxide, $\Delta H_f(\text{MgO})$.



Note: If you were unable to calculate the enthalpy changes, assume that the value of $\Delta H_{\text{rxn } 1}$ is -457 kJ mol^{-1} and the value $\Delta H_{\text{rxn } 2}$ is -141 kJ mol^{-1} .



By Hess's law,

$$\begin{aligned}\Delta H_f(\text{MgO}) &= \Delta H_{\text{rxn } 1} + (-286) - (\Delta H_{\text{rxn } 2}) \\ &= -464 + (-286) - (-117) \text{ (or } -457 + (-286) - (-141)) \\ &= -633 \text{ (or } -602) \text{ kJ mol}^{-1} \text{ [1] with sign}\end{aligned}$$

$$\Delta H_f(\text{MgO}) = -633 \text{ (3 s.f. or exact value) kJ mol}^{-1} \text{ [2]}$$

- (d) Two methods, **Method 1** and **Method 2**, have been used to determine the maximum temperature change, ΔT , of an exothermic reaction. Which method would give a more accurate ΔT ? Explain your answer.

ΔT obtained by Method 1 is more accurate as a cooling correction has been done by the extrapolating the cooling curve to the time of mixing to compensate/account for the heat loss to the surroundings. [1]

[1]

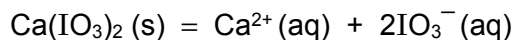
Bonus [1]: shows workings in all calculations + gives all final answers to 3 s.f. in 1(c), 1(d), 2(a)(iii), 2(b)(iii) and 2(c).

[Total: 16]

3 Planning

The solubility of calcium iodate(V), $\text{Ca}(\text{IO}_3)_2$, at 20 °C, is approximately 2.4 g dm^{-3} .

When solid calcium iodate(V) is added to water, a small amount dissolves to form a saturated solution, establishing an equilibrium between the undissolved salt and its aqueous ions.

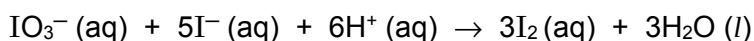


The equilibrium constant for the above solubility equilibrium, K_{sp} , is also known as the solubility product of calcium iodate.

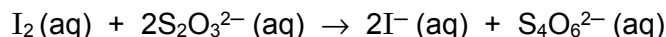
$$K_{\text{sp}} = [\text{Ca}^{2+}(\text{aq})] [\text{IO}_3^-(\text{aq})]^2$$

This solubility product can be found by determining the equilibrium concentration of IO_3^- ions in a saturated solution of calcium iodate.

The exact concentration of IO_3^- ions is determined by titration. Excess aqueous KI and aqueous H^+ is first added to a sample of saturated calcium iodate solution to liberate iodine.



The iodine liberated in the resulting mixture is then titrated with sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$ of known concentration.



- (a) Using the information given above, you are required to write a plan to determine the solubility product, K_{sp} , of calcium iodate, $\text{Ca}(\text{IO}_3)_2$, at 20 °C.

You may assume that you are provided with:

- solid sodium thiosulfate crystals, $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$ ($M_r = 248.2$)
- solid calcium iodate, $\text{Ca}(\text{IO}_3)_2$
- aqueous potassium iodide, KI, of about 0.2 mol dm^{-3}
- aqueous hydrochloric acid, HCl, of about 1 mol dm^{-3}
- starch indicator
- any other required apparatus normally found in a college laboratory.

Your plan should include details of, including quantities:

- the preparation of 250.0 cm^3 of $0.075 \text{ mol dm}^{-3}$ aqueous $\text{Na}_2\text{S}_2\text{O}_3$;
- the preparation of about 100 cm^3 of a saturated solution of calcium iodate, $\text{Ca}(\text{IO}_3)_2$ at 20 °C;
- the essential details of the titration process;
- an outline of how you would use your mean titre value to determine the solubility product of $\text{Ca}(\text{IO}_3)_2$.

In your calculations, you should let $V \text{ cm}^3$ be your mean titre and express your final mathematical expression in terms of V .

To prepare 250.0 cm³ of 0.075 mol dm⁻³ Na₂S₂O₃(aq):

$$\text{Mass of Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O required} = \frac{250}{1000} \times 0.075 \times 248.2 = \text{*4.65 g}$$

1. Weigh accurately *4.65 g of solid Na₂S₂O₃·5H₂O into a weighing bottle using an electronic weighing balance.

[1] weigh correct mass of solid used in 2-3 d.p. (accept a range)

2. Transfer all the weighed solid into a small beaker and dissolve the solid completely with about 50 cm³ deionised water.

3. Using a filter funnel, carefully transfer the solution and all washings into a 250.0 cm³ graduated flask. Make up to the graduated mark with deionised water. Stopper and shake the flask well to obtain a homogeneous solution.

[1] accurately prepares 250 cm³ of standard soln with complete transfer (no loss of S₂O₃²⁻)

To prepare 100 cm³ of Ca(IO₃)₂ saturated solution at 20 °C:

1. Use a 100 cm³ measuring cylinder to transfer 100 cm³ of deionised water into a 250 cm³ conical flask.

2. Using a spatula, add a few tips of solid Ca(IO₃)₂ into the conical flask. Stopper the flask and shake the flask for a few minutes. Keep adding more solid Ca(IO₃)₂, with shaking after each addition, until some Ca(IO₃)₂ solids are left undissolved.

To ensure that the solution is saturated, shake the flask at intervals and leave the conical flask in a *thermostat controlled water bath set at 20 °C for some time. There must be some solids left undissolved.

[1] correctly prepares a saturated soln of Ca(IO₃)₂ at 20 °C (with *) (allow stirring or swirling in place of shaking)

3. To remove undissolved solids, filter the saturated solution into a clean, dry conical flask using a dry filter funnel and a piece of dry filter paper. [1] dry filtration

Titration procedure:

1. Fill a burette with 0.075 mol dm⁻³ Na₂S₂O₃(aq).

2. Pipette 25.0 cm³ of the saturated solution into a conical flask.

3. Use a 10 cm³ measuring cylinder to add to the flask 5 cm³ of HCl(aq) and 10 cm³ of KI(aq).

[1] accurately prepares titrant and analyte with min 10 cm³ of KI & 5 cm³ of H⁺

4. Titrate the liberated I₂ in the mixture with Na₂S₂O₃(aq) from the burette until the solution turns pale yellow. Then add about 1 cm³ of starch indicator and continue the titration until the dark blue-black colour just disappears/turns colourless at the end-point.

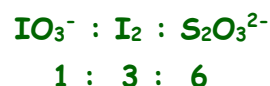
[1] correct titration procedure using starch (allow a few drops)

5. Repeat the titration to obtain two consistent titres within 0.10 cm³ in difference.

To calculate K_{sp} of $\text{Ca}(\text{IO}_3)_2$:

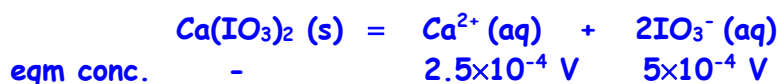
$$\text{Amount of } \text{S}_2\text{O}_3^{2-} \text{ required} = \frac{V}{1000} \times 0.075 = 7.5 \times 10^{-5} \text{ V mol}$$

$$\begin{aligned} \text{Amount of } \text{IO}_3^- \text{ in } 25.0 \text{ cm}^3 \text{ saturated solution} \\ = \frac{1}{6} \times 7.5 \times 10^{-5} \text{ V} = 1.25 \times 10^{-5} \text{ V mol} \end{aligned}$$



[1]

$$[\text{IO}_3^-] \text{ in saturated solution} = 1.25 \times 10^{-5} \text{ V} \times \frac{1000}{25.0} = 5 \times 10^{-4} \text{ V mol dm}^{-3}$$



$$\begin{aligned} K_{sp} (\text{Ca}(\text{IO}_3)_2) &= [\text{Ca}^{2+}(\text{aq})] [\text{IO}_3^-(\text{aq})]^2 \\ &= (2.5 \times 10^{-4} \text{ V})(5 \times 10^{-4} \text{ V})^2 \\ &= 6.25 \times 10^{-11} \text{ V}^3 \text{ mol}^3 \text{ dm}^{-9} \end{aligned}$$

[1]

[8]

- (b) The solubility of calcium iodate(V), $\text{Ca}(\text{IO}_3)_2$, at 20°C , is approximately 2.4 g dm^{-3} . Justify, with calculations, that the chosen concentration, $0.075 \text{ mol dm}^{-3}$, of aqueous $\text{Na}_2\text{S}_2\text{O}_3$ solution is appropriate. [M_r of $\text{Ca}(\text{IO}_3)_2 = 389.9$]

$$\text{Expected } [\text{IO}_3^-] \text{ in the saturated solution} = \frac{2.4}{389.9} \times 2 = 0.0123 \text{ mol dm}^{-3}$$

[1]

$$\text{Amount of } \text{IO}_3^- \text{ in } 25.0 \text{ cm}^3 = \frac{25.0}{1000} \times 0.0123 = 3.08 \times 10^{-4} \text{ mol}$$

$$\text{Amount of } \text{S}_2\text{O}_3^{2-} \text{ required} = 3.08 \times 10^{-4} \times 6 = 1.85 \times 10^{-3} \text{ mol}$$

Expected titre when $0.075 \text{ mol dm}^{-3}$ $\text{Na}_2\text{S}_2\text{O}_3$ is used

$$= \frac{1.85 \times 10^{-3}}{0.075} = 0.0246 \text{ dm}^3$$

= 24.6 cm^3 which is within the capacity of a burette.

[1]

[2]

- (c) The experiment described in your plan in (a) is repeated using 0.1 mol dm^{-3} aqueous $\text{Ca}(\text{NO}_3)_2$ solution, instead of deionised water, to prepare a saturated solution of calcium iodate(V).

State and explain how, if at all, the titre values and calculated K_{sp} would be expected to differ from that obtained in (a).

[Assume that both the experiments were carried out under the same conditions.]



Presence of common ion Ca^{2+} from $\text{Ca}(\text{NO}_3)_2$ causes the position of equilibrium (1) to shift left (or the solubility of $\text{Ca}(\text{NO}_3)_2$ in 0.10 mol dm^{-3} $\text{Ca}(\text{NO}_3)_2$ is lower) and hence the titre values will be smaller due to lower $[\text{IO}_3^-]$ in saturated solution. [1]

However, since the temperature is kept constant, the calculated K_{sp} value would remain the same/unchanged. [1]

Award [1] for correct conclusions: smaller titre and constant K_{sp} value

[2]

[Total: 12]

4 Qualitative Analysis

- (a) You are provided with solid **FA 6** which contains $\text{S}_2\text{O}_8^{2-}$ anion and one cation from the ions listed in the *Qualitative Analysis Notes*.

You are to perform the tests described in Table 4.1 to deduce the nature of **FA 6** and to identify the cation present in **FA 6**.

Record your observations in Table 4.1. Your answers should include

- details of colour changes and precipitates formed,
- the identity of any gases evolved, and details of the test used to identify each one.

You should indicate clearly at what stage in a test a change occurs.

Marks are not given for chemical equations.

No additional or confirmatory tests for ions present should be attempted.

Table 4.1

Tests	Observations
(i) To a spatula measure of FA 6 in a boiling tube, add 1-2 cm depth of dilute hydrochloric acid and heat cautiously. Retain the solution for test (ii).	• <u>Pungent, yellowish green Cl_2 gas</u> (✓) • <u>bleaches a damp blue litmus paper</u> (✓) • <u>pale yellow solution obtained.</u> {FA 8 oxidises Cl^- to Cl_2}
(ii) To the resulting solution from test (i), add barium nitrate(V) solution.	• <u>White ppt.</u> (✓) insoluble in H^+ {BaSO_4 ppt. $\Rightarrow$ SO_4^{2-} formed}
(iii) To a spatula measure of FA 6 in a test tube, add 1-2 cm depth of freshly prepared solution of iron(II) sulfate solution and warm.	• <u>Pale green Fe^{2+} solution turns yellow</u> • <u>Upon warming, solution turns brown/red-brown/orange-brown/orange.</u> (✓) {FA 8 oxidises Fe^{2+} to Fe^{3+}}
(iv) Place a spatula measure of FA 6 in a boiling tube and add dilute nitric acid to dissolve the solid. Then add 1 cm depth of manganese(II) sulfate solution and 4 drops of silver nitrate(V) solution to act as a catalyst. Heat cautiously to bring the mixture to boiling.	• <u>Upon heating, pale pink Mn^{2+} solution turns brown</u> (✓) <u>then to a black ppt./purple solution/black suspension.</u> (✓) {FA 8 oxidises Mn^{2+} to Mn^{3+} then to $\text{MnO}_2/\text{MnO}_4^-$}
(v) To a spatula measure of FA 6 in a boiling tube, add 1 cm depth of aqueous sodium hydroxide. Heat cautiously.	• <u>No ppt.</u> (✓) • <u>Upon heating, colourless, pungent NH_3 gas</u> (✓) • <u>turns damp red litmus paper blue</u> (✓) {NH_4^+ present}

[5]

8-9(✓) [5], 6-7(✓) [4], 4-5(✓) [3], 2-3(✓) [2], 1(✓) [1]

- (b) (i) State, with supporting evidence, the nature of FA 6.

nature of FA 6: oxidising agent [1]

evidence: FA 6 oxidises Fe^{2+} in test (iii) to Fe^{3+} (or Cl^- in test (i) to Cl_2 or Mn^{2+} in test (iv) to $\text{Mn}^{3+}/\text{MnO}_2/\text{MnO}_4^-$) [1]

- (ii) Identify the cation in FA 6.

cation: NH_4^+ [1]

- (iii) Write the ionic equation, including state symbols, for the reaction that occurs in test (i).



- (c) You are provided with an organic solution FA 7 which contains one functional group.

Care: FA 7 is flammable. Do not use Bunsen burner for heating. Use the hot water provided.

FA 7 gives a positive test with 2,4-dinitrophenylhydrazine.

Devise **two** other confirmatory tests using the **bench reagents provided** to identify the functional group present in FA 7.

Carry out the tests and record details of the tests performed and observations made in Table 4.2.

Table 4.2

Confirmatory Tests	Observations
To 1 cm depth of FA 7 in a test-tube, <u>add 1-2 cm depth of dilute $\text{H}_2\text{SO}_4(\text{aq})$ and 1-2 drop of $\text{KMnO}_4(\text{aq})$. Heat in hot water-bath.</u> [1]	<u>Purple KMnO_4 decolourised.</u> [1]
To 1 cm depth of FA 7 in a test-tube, <u>add 1 cm depth of Fehling's solution. Heat in hot water-bath.</u> [1]	<u>Red-brown / brick-red ppt. formed.</u> [1]

Functional group present in FA 7: aliphatic aldehyde [1] [5]

[Total: 14]

Qualitative Analysis Notes**[ppt. = precipitate]****(a) Reactions of aqueous cations**

cation	reaction with	
	NaOH(aq)	NH ₃ (aq)
aluminium, Al ³⁺ (aq)	white ppt. soluble in excess	white ppt. insoluble in excess
ammonium, NH ₄ ⁺ (aq)	ammonia produced on heating	–
barium, Ba ²⁺ (aq)	no ppt. (if reagents are pure)	no ppt.
calcium, Ca ²⁺ (aq)	white ppt. with high [Ca ²⁺ (aq)]	no. ppt.
chromium(III), Cr ³⁺ (aq)	grey-green ppt. soluble in excess giving dark green solution	grey-green ppt. insoluble in excess
copper(II), Cu ²⁺ (aq)	pale blue ppt. insoluble in excess	blue ppt. soluble in excess giving dark blue solution
iron(II), Fe ²⁺ (aq)	green ppt., turning brown on contact with air insoluble in excess	green ppt., turning brown on contact with air insoluble in excess
iron(III), Fe ³⁺ (aq)	red-brown ppt. insoluble in excess	red-brown ppt. insoluble in excess
magnesium, Mg ²⁺ (aq)	white ppt. insoluble in excess	white ppt. insoluble in excess
manganese(II), Mn ²⁺ (aq)	off-white ppt., rapidly turning brown on contact with air insoluble in excess	off-white ppt., rapidly turning brown on contact with air insoluble in excess
zinc, Zn ²⁺ (aq)	white ppt. soluble in excess	white ppt. soluble in excess

(b) Reactions of anions

<i>anion</i>	<i>reaction</i>
carbonate, CO_3^{2-}	CO_2 liberated by dilute acids
chloride, $\text{Cl}^-(\text{aq})$	gives white ppt. with $\text{Ag}^+(\text{aq})$ (soluble in $\text{NH}_3(\text{aq})$)
bromide, $\text{Br}^-(\text{aq})$	gives pale cream ppt. with $\text{Ag}^+(\text{aq})$ (partially soluble in $\text{NH}_3(\text{aq})$)
iodide, $\text{I}^-(\text{aq})$	gives yellow ppt. with $\text{Ag}^+(\text{aq})$ (insoluble in $\text{NH}_3(\text{aq})$)
nitrate, $\text{NO}_3^-(\text{aq})$	NH_3 liberated on heating with $\text{OH}^-(\text{aq})$ and Al foil
nitrite, $\text{NO}_2^-(\text{aq})$	NH_3 liberated on heating with $\text{OH}^-(\text{aq})$ and Al foil; NO liberated by dilute acids (colourless $\text{NO} \rightarrow$ (pale) brown NO_2 in air)
sulfate, $\text{SO}_4^{2-}(\text{aq})$	gives white ppt. with $\text{Ba}^{2+}(\text{aq})$ (insoluble in excess dilute strong acids)
sulfite, $\text{SO}_3^{2-}(\text{aq})$	SO_2 liberated with dilute acids; gives white ppt. with $\text{Ba}^{2+}(\text{aq})$ (soluble in excess dilute strong acids)

(c) Tests for gases

<i>gas</i>	<i>test and test results</i>
ammonia, NH_3	turns damp red litmus paper blue
carbon dioxide, CO_2	gives a white ppt. with limewater (ppt. dissolves with excess CO_2)
chlorine, Cl_2	bleaches damp litmus paper
hydrogen, H_2	"pops" with a lighted splint
oxygen, O_2	relights a glowing splint
sulfur dioxide, SO_2	turns aqueous acidified potassium manganate(VII) from purple to colourless

(d) Colour of halogens

<i>halogen</i>	<i>colour of element</i>	<i>colour in aqueous solution</i>	<i>colour in hexane</i>
chlorine, Cl_2	greenish yellow gas	pale yellow	pale yellow
bromine, Br_2	reddish brown gas / liquid	orange	orange-red
iodine, I_2	black solid / purple gas	brown	purple

2018 Prelim Exam Paper 4 Chemical list:

Label	Identity	Preparation	per candidate
FA 1	0.1 mol dm ⁻³ hydrochloric acid	Dilute 50 cm ³ of 2.0 mol dm ⁻³ HCl to 1 dm ³ with deionised water.	200 cm ³
FA 2	0.1 mol dm ⁻³ sodium hydroxide	Dissolve 4.00g of NaOH [C] in each dm ³ of solution.	200 cm ³
FA 3	Mg ribbon	0.11 – 0.13 g of Mg ribbon supplied in a stoppered weighing bottle (*cannot exceed 0.135 g) <u>Note</u> : sand Mg ribbons and cut into ≈ 1.5 - 2 cm length before weighing	1 bottle
FA 5	2 mol dm ⁻³ hydrochloric acid	Dilute 170 cm ³ of concentrated (35-37%; ≈ 11 mol dm ⁻³) hydrochloric acid [C] to 1 dm ³ with deionised water	150 cm ³
FA 6	solid ammonium persulfate, (NH ₄) ₂ S ₂ O ₈	About 3 g of solid (NH ₄) ₂ S ₂ O ₈ supplied in a stoppered weighing bottle	1 bottle
FA 7	Ethanal	10% (v/v) aqueous solution of ethanal (acetaldehyde) To reduce the possibility of the presence of acid, a <u>fresh batch</u> of ethanal should be used.	1 bottle (8-10 cm ³)
Methyl orange indicator			1 bottle

Central Supplies:

aqueous iron(II) sulfate, FeSO ₄ (freshly prepared)	0.5 mol dm ⁻³ FeSO ₄ (aq) Dissolve 140 g of FeSO ₄ .7H ₂ O in 1 dm ³ of water containing 7 cm ³ of concentrated H ₂ SO ₄ .	2 bottles per lab (about 5 cm ³ per student)
aqueous manganese(II) sulfate, MnSO ₄	0.1 mol dm ⁻³ MnSO ₄ (aq) Dissolve 16.9 g of MnSO ₄ .H ₂ O in 1 dm ³ of water	2 bottles per lab (about 5 cm ³ per student)
Fehling's solution		2 bottles per lab (about 5 cm ³ per student)
Hot water		200 cm ³ per student
Small white labels (Teacher's bench)		
Electronic Weighing balance		4 per lab

Student's QA reagents:

aq. potassium manganate(VII), KMnO_4	Standard QA reagents concentrations	1 bottle
aq. silver nitrate(V), AgNO_3		1 bottle
aq. barium nitrate(V), $\text{Ba}(\text{NO}_3)_2$		1 bottle
aq. sodium hydroxide, NaOH		1 bottle
dilute hydrochloric acid, HCl		1 bottle
dilute sulfuric acid, H_2SO_4		1 bottle
dilute nitric acid, HNO_3		1 bottle
aqueous ammonia, NH_3		1 bottle
limewater (freshly prepared)		1 bottle
Red litmus paper		4 pieces
Blue litmus paper		4 pieces
Filter paper strip		2 pieces
Wooden splint		2 pieces

Student's apparatus:		per candidate
1	50.00 cm^3 burette	1
2	Burette stand and clamp	1
3	25.0 cm^3 pipette	1
4	Pipette filler	1
5	250 cm^3 conical flask	2
6	Filter funnel	1
7	White tile	1
8	50 cm^3 measuring cylinder	1
9	Styrofoam cup	2 (supported in a big beaker)
10	Thermometer ($-5\text{ }^\circ\text{C}$ to $+50\text{ }^\circ\text{C}$) 0.2 $^\circ\text{C}$ interval	1
11	Stop-watch	1
12	Plastic dropper	2
13	100 cm^3 beaker	1 (for burette waste)
14	200 cm^3 beaker	1 (for hot water bath)
15	Wash bottle with deionised water	1
16	Test-tube rack	1
17	Test-tube	6 } inside a Ziploc bag
18	Boiling tube	
19	Bunsen Burner	1
20	Test-tube holder	1
21	Glass rod	1
22	Test-tube brush	1
23	Delivery tube (for test-tube)	1
24	Delivery tube (for boiling tube)	1
25	Plastic spatula (small measure)	1
26	Paper Towel	2 pieces